

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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36. Recognise watch from the given image by general Object recognition using OpenCV.

AIM:

Recognise watch from the given image by general Object recognition

PROGRAM:

```
import cv2

# Read the image
img = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp3.png")

if img is None:
    print("Error: Image not found!")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Reduce noise
    blur = cv2.GaussianBlur(gray, (5, 5), 0)

    # Detect edges
    edges = cv2.Canny(blur, 50, 150)

    # Find contours
    contours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)

    # Draw bounding box and label
    for contour in contours:
        area = cv2.contourArea(contour)

        # Ignore very small contours
        if area > 1000:
            x, y, w, h = cv2.boundingRect(contour)

            cv2.rectangle(img, (x, y), (x + w, y + h), (0, 255, 0), 2)

            cv2.putText(img, "Watch",
```

```
(x, y - 10),  
cv2.FONT_HERSHEY_SIMPLEX,  
0.8,  
(0, 255, 0),  
2)  
  
# Save output  
cv2.imwrite(  
    r"C:\Users\Susmitha\Downloads\ITA0510 Lab\Recognized_Watch.jpg",  
    img  
)  
  
# Display output  
cv2.imshow("Recognized Watch", img)  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

INPUT:



RESULT:

Thus the program is executed successfully.